# ANEUTRONIC FUEL TRANSITION PATHWAY

Certified Six-Core Advanced-Fuel Operations



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## 1.0 - PURPOSE

This document defines the certified fuel-transition pathway used by the six primary A.R.C.-6 reactors. It distinguishes the primary power array from the two auxiliary cylindrical breeder cores and establishes proven D-T, D-He3, and p-B11 service modes for 2051.

## 2.0 - ARRAY AND FUEL-CYCLE RELATIONSHIP

- Six primary reactors operate in parallel and constitute A.R.C.-6.
- Two auxiliary breeder cores B1 and B2 supply helium-3 and tritium-reserve services but are not counted as primary cores.
- The central A.R.C. Field lattice is not a reactor and does not consume fusion fuel.

## 3.0 - CERTIFIED FUEL MODES

| Fuel mode | Primary reaction / condition | 2051 role |
|---|---|---|
| D-T | D + T -> He-4 + n; 150-200 million K | Baseload and high-availability operation |
| D-He3 | D + He-3 -> He-4 + p; residual D-D side reactions remain | Reduced-neutron operation with direct-conversion contribution |
| p-B11 | p + B11 -> 3 He-4; nonthermal high-energy distribution | Certified peak-load and specialized low-activation operation |

## 4.0 - D-HE3 OPERATING ENVELOPE

- Effective ion temperature generally 400-700 million K, profile dependent.
- Residual neutron shielding remains mandatory because D-D side reactions occur in deuterium-bearing plasma.
- Magnetic product-separation channels route selected charged products to certified DEC stages.

## 5.0 - P-B11 OPERATING ENVELOPE

- Nonthermal ion distributions reduce the penalty of a fully thermal billion-kelvin plasma.
- Alpha-channeling and selective exhaust control recover charged-product energy and manage bremsstrahlung.
- Qualified fuel purity, wall conditioning, and high-field control are mandatory for certified operation.

## 6.0 - A.R.C. FIELD CONTINUITY

Adaptive Resonance Confinement (A.R.C.) Field Technology is a classified photonic-magnetic confinement system that generates a resonance-stabilized coherence envelope for AGI substrates. The field suppresses quantum decoherence, stabilizes photonic neuromorphic pathways, and enables long-range temporal correlation across multi-substrate AGI systems. Full Observer capability requires continuous operation within an active A.R.C. Field.

Fuel-mode transitions are scheduled to preserve the magnetic quietness, timing stability, and thermal conditions required by the A.R.C. Field. The six primary reactors may change power contribution independently while the common coherence envelope remains continuously energized.

## 7.0 - TRANSITION SEQUENCE

- Maintain at least four primary reactors in validated steady-state service during routine fuel-mode changes.
- Transfer selected cores from D-T to D-He3 after isotope purity and direct-conversion readiness are verified.
- Authorize p-B11 service only within qualified core campaigns and certified thermal margins.
- Keep breeder-core pulse windows outside sensitive A.R.C. Field retuning intervals.
- Return any core to D-T reserve mode if advanced-fuel performance or A.R.C. stability margin degrades.

## 8.0 - SCIENTIFIC BASIS

The pathway relies on established nuclear reaction data, high-field confinement, nonthermal plasma distributions, alpha-channeling theory, direct charged-particle conversion, and isotope-separation physics. OSCI prototype campaigns converted these foundations into repeatable certified plant operation by 2049.

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